Mathematics for Intelligent Systems 3

23MAT204

Linear Programming (LP)
Simplex Method

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Disclaimer

These slides may contain unintentional errors.
I would greatly appreciate it if you could point them out.
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References

Main references:

- *Convex Optimization*, Stephen Boyd and Lieven Vandenberghe
  https://web.stanford.edu/~boyd/cvxbook/bv_cvxbook.pdf
- *Operations Research*, Taha, H. A. : An Introduction, 10th ed., Pearson (2017)

Key Ideas: Simplex Method

- Solves linear programming models by iteratively improving feasible solutions.
- Uses slack, surplus, and artificial variables to standardize constraints.
- Each step represented by a tableau; pivots exchange basic/non-basic variables.

Flow chart

Step 1: Standard Form

Express the L.P.P. in standard form

Step 2: Initial Simplex Tableau

- Write the initial simplex tableau.
- A tableau arranges all variables, constraints, and coefficients for iterative row operations.
- Identify basic (in basis) and non-basic (set to zero) variables.

Step 3: Compute Row Values

- Calculate Z_j: The sum of product of coefficients and values for basic variables.
- Compute $c_j - Z_j$: Indicates potential for improvement if variable enters basis.

Step 4: Identify Pivot Column

- Problem type:
  - **Minimization:** Choose most negative $c_j - Z_j$.
  - **Maximization:** Choose largest positive $c_j - Z_j$.
- Pivot column indicates variable to enter basis.

Step 5: Pivot Row Element

- Calculate ratios b_i/a_{ij} (right-hand side divided by pivot column element).
- Select key row with minimum non-negative ratio; the intersection gives pivot element.
- If all ratios are negative or infinite, **solution is unbounded**.

Step 6: Pivot and Row Operations

- Divide pivot row by pivot element to make pivot position "1".
- Update other rows to make pivot column entries "0".
- Results in new tableau; update basis accordingly.

Step 7: Iteration Optimality

- Recompute Z_j and $c_j - Z_j$ values.
- Check for optimality:
  - **Minimization:** Are all $c_j - Z_j \geq 0$?
  - **Maximization:** Are all $c_j - Z_j \leq 0$?
- If YES, solution is optimal; if NO, repeat pivot process.

Special Cases

Unbounded Solution

If pivot column has all negative or zero coefficients, feasible solution does not exist and problem is unbounded.

Infeasible Solution

If artificial variables are still in the basis at optimality, solution is infeasible.

Simplex Method: Example Solution

Problem Statement:

$$\begin{aligned} \text{Maximize: } & Z = 3x_1 + 4x_2 \\ \text{Subject to: } & x_1 + x_2 \leq 450 \quad \text{(Machine } M_1 \text{ constraint)} \\ & 2x_1 + x_2 \leq 600 \quad \text{(Machine } M_2 \text{ constraint)} \\ & x_1, x_2 \geq 0 \end{aligned}$$

Step 1: Standard Form

Add slack variables s_1 and s_2 to convert inequalities to equalities:

$$\begin{aligned} \text{Maximize: } & Z = 3x_1 + 4x_2 + 0s_1 + 0s_2 \\ \text{Subject to: } & x_1 + x_2 + s_1 = 450 \\ & 2x_1 + x_2 + s_2 = 600 \\ & x_1, x_2, s_1, s_2 \geq 0 \end{aligned}$$

Step 2: Initial Simplex Tableau

c_j		3	4	0	0	
C_B	Basis	x_1	x_2	s_1	s_2	b
0	s_1	1	1	1	0	450
0	s_2	2	1	0	1	600

The solution is

$$x_1 = 0, \quad x_2 = 0.$$

Basis

The basic variables are

$$s_1 = 450, \quad s_2 = 600,$$

representing 450 minutes of unused M1 time and 600 minutes of unused M2 time.

Profit

The current total profit is \$0.

Step 3: Optimality Test

c_j		3	4	0	0		
C_B	Basis	x_1	x_2	s_1	s_2	b	θ
0	s_1	1	**(1)**	1	0	450	$450 / 1 = 450$
0	s_2	2	1	0	1	600	$600 / 1 = 600$
Z_j		0	0	0	0	0	
$c_j - Z_j$		3	**4**	0	0		

- $c_j - Z_j$ **Row**: Shows the net profit gained per unit. Since the values for x_1 (3) and x_2 (4) are positive, the solution is **not optimal**.
- **Entering Variable**: x_2 is chosen because it has the highest positive value (4) in the $c_j - Z_j$ row. This identifies the **key column**.

- **Leaving Variable**: The minimum ratio test is performed by dividing the 'b' column by the key column values:

$$\min\left(\frac{450}{1}, \frac{600}{1}\right) = 450$$

  The minimum ratio of 450 determines that s_1 leaves the basis. This identifies the **key row**.
- **Key Element**: The element at the intersection of the key column and key row is **(1)**.

Step 4: Row Operations and New Tableau

Pivot on the element (1) at the intersection of the key row and key column.

c_j		3	4	0	0	
C_B	Basis	x_1	x_2	s_1	s_2	b
4	x_2	1	1	1	0	450
0	s_2	1	0	-1	1	150

Step 5: Final Optimality Test

c_j		3	4	0	0	
C_B	Basis	x_1	x_2	s_1	s_2	b
4	x_2	1	1	1	0	450
0	s_2	1	0	-1	1	150
Z_j		4	4	4	0	1800
$c_j - Z_j$		-1	0	-4	0	

- **New Solution**: The basic variables are now $x_2 = 450$ and $s_2 = 150$. We are producing 450 units of product 2, leaving 150 minutes of idle time on Machine M2.
- **New Profit**: The total profit is now \$1,800.
- **Optimality Check**: All values in the $c_j - Z_j$ row are zero or negative. This means the solution is **optimal**.

Optimal Solution

All $c_j - Z_j$ values are zero or negative, so the solution is optimal.

$$x_1 = 0$$
$$x_2 = 450$$
$$Z_{max} = 1800$$

Belt Production Problem

A company makes two kinds of leather belts, belt A and belt B. Belt A is a high quality belt and belt B is of lower quality. The respective profits are Rs 4 and Rs 3 per belt. The production of each of type A requires twice as much time as a belt of type B, and if all belts were of type B, the company could make 1,000 belts per day. The supply of leather is sufficient for only 800 belts per day (both A and B combined). Belt A requires a fancy buckle and only 400 of these are available per day. There are only 700 buckles a day available for belt B.
What should be the daily production of each type of belt? Formulate this problem as an LP model and solve it using the simplex method.

Formulating problems as LP

A company makes two kinds of leather belts:

- Belt A (high quality): Profit = Rs. 4 per unit
- Belt B (lower quality): Profit = Rs. 3 per unit

Decision Variables

$$x_1 = \text{number of Belt A produced per day},$$

$$x_2 = \text{number of Belt B produced per day}.$$

Objective Function Maximize profit:

$$\text{Maximize } Z = 4x_1 + 3x_2$$

Constraints

1. **Time constraint:** Belt A takes twice as much time as Belt B. If only B is produced, 1000 belts can be made. Hence,

$$2x_1 + x_2 \leq 1000.$$

2. **Leather supply constraint:** At most 800 belts (A and B combined):

$$x_1 + x_2 \leq 800.$$

3. **Belt A buckle constraint:** Only 400 fancy buckles for belt A:

$$x_1 \leq 400.$$

4. **Belt B buckle constraint:** Only 700 buckles for belt B:

$$x_2 \leq 700.$$

5. **Non-negativity:**

$$x_1, x_2 \geq 0.$$

The LP model:

$$
\begin{aligned}
\text{Maximize } & Z = 4x_1 + 3x_2 \\
\text{subject to } & 2x_1 + x_2 \leq 1000, \\
& x_1 + x_2 \leq 800, \\
& x_1 \leq 400, \\
& x_2 \leq 700, \\
& x_1, x_2 \geq 0.
\end{aligned}
$$

Optimal Solution

$$\boxed{x_1 = 200, \quad x_2 = 600, \quad Z_{\max} = Rs.2600}$$

Practice problem

$$
\begin{aligned}
\text{Maximize } Z &= 2x_1 + 5x_2 \\
\text{subject to } x_1 &+ 4x_2 \leq 24, \\
3x_1 &+ x_2 \leq 21, \\
x_1 &+ x_2 \leq 9, \\
x_1, x_2 &\geq 0.
\end{aligned}
$$

Optimal Solution

$$\boxed{x_1 = 4, \quad x_2 = 5, \quad Z_{\max} = 33}$$